

Panel B operationalizes *bridge accuracy* across 25 validation conditions. For each condition r :

1. Compute the truth θ_s^{true} from linked data.
2. Erase the pairings; retain only the unlinked marginals F_U and F_V .
3. Re-run the copula bridge (STEP 3) to obtain $\hat{\theta}_s^{\text{bridge}}$.
4. Record the bridge error:

$$\text{Bridge error}_r = \hat{\theta}_s^{\text{bridge}} - \theta_s^{\text{true}}.$$

The scatter plot shows *truth vs. bridge estimate* for each condition. Points near the identity line indicate high accuracy. The amber band marks ± 2 SGPC points from perfect agreement.

Accuracy is summarized by center displacement, not by interval width. The relevant metrics are mean absolute error (MAE = 0.42), mean signed error (-0.01), and RMSE (0.55), all on the 0–100 SGPC scale.

This validation answers: *how far does the unlinked bridge move us from the truth that linked data would provide?*²